Matrices of Linear Transformations

Geometric Algorithms
Lecture 8

CAS CS 132

Keywords

matrix of a linear transformation

standard basis vectors (standard coordinate vectors)

2D linear transformations

the unit square

one-to-one

onto

Motivating Question

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$$\vec{x} \mapsto A\vec{x}$$

We know that matrix transformations are linear transformations

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Are there any other kinds of linear transformations?

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Are there any other kinds of linear transformations?

NO

Matrix of a Linear Transformation

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Linear transformations are **exactly** matrix transformations

Fundamental Concern

Given a linear transformation T, how do we find the matrix A such that $T(\mathbf{v}) = A\mathbf{v}$?

Thought Experiment

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This is basically linear algebraic battleship

Recall: Calculating $A\mathbf{v}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ \vdots \\ ? \end{bmatrix}$$

$$b_1 = a_{11}v_1 + a_{12}v_2 + \ldots + a_{1n}v_n = \sum_{i=1}^{n} a_{1i}v_i$$

Recall: Matrix-Vector Multiplication

Definition. Given a $(m \times n)$ matrix A with columns $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$, and a vector $\mathbf{v}$ in $\mathbb{R}^n$, we define

$$A\mathbf{v} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \ldots \quad \mathbf{a}_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{a}_1 + v_2\mathbf{a}_2 + \ldots v_n\mathbf{a}_n$$

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$A\mathbf{v}$ is a linear combination of the columns of A with weights given by $\mathbf{v}$

Isolating a_{11}

$$b_1 = \boxed{a_{11}\underset{1}{v_1} + a_{12}\underset{0}{v_2} + \ldots + a_{1n}\underset{0}{v_n}} = \sum_{i=1}^{n} a_{1i}v_i$$

$$A\begin{bmatrix}1\\0\\\vdots\\0\end{bmatrix} = \underbrace{\begin{bmatrix}a_{11}\\a_{21}\\\vdots\\a_{m1}\end{bmatrix}}_{\text{first col. of } A}$$

The Takeaway

We can learn the first column of the matrix

implementing T by looking at $T\left(\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}\right)$

Matrix of a Linear Transformation

Standard Basis

Definition. The n**–*dimensional standard basis vectors*** (or standard coordinate vectors) are the vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$ where

$$\mathbf{e}_i = \begin{array}{c} \\ \end{array} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \begin{array}{c} 1 \\ 2 \\ \vdots \\ i-1 \\ i \\ i+1 \\ \vdots \\ n-1 \\ n \end{array}$$

Standard Basis

Definition (Alternative). The n–dimensional standard basis vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are the columns of the $n \times n$ identity matrix

$$I = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \cdots \quad \mathbf{e}_n]$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Standard Basis and the Matrix Equation

$1 \ \vec{a}_i$

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n]\,\mathbf{e}_i = \mathbf{a}_i$$

$$\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

The standard basis vectors gives us a way to "look into" a matrix

Standard Basis and Vector Coordinates

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \ldots + v_n\mathbf{e}_n$$

Column vectors describe how to write a vector as a linear combination of the standard basis

Standard Basis and Linear Transformations

Theorem. For any linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$, the matrix

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix}$$

is the <u>unique</u> matrix such that $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v}$ in $\mathbb{R}^n$

Formally

$$T(\mathbf{v}) = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix} \mathbf{v}$$

$$\vec{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = v_1 \vec{e}_n + \dots v_n \vec{e}_n$$

$$T(\vec{v}) = T(\vec{v}_1 \vec{e}_1 + \dots + v_n \vec{e}_n)$$

$$= v_1 T(\vec{e}_1) + \dots + v_n T(\vec{e}_n)$$

$$= \begin{bmatrix} T(\vec{e}_1) & \dots & T(\vec{e}_n) \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

How To: Matrices of Linear Transformations

Question. Find the matrix which implements the transformation $T : \mathbb{R}^n \to \mathbb{R}^m$

Solution. Determine the images of standard basis under T. Then write down

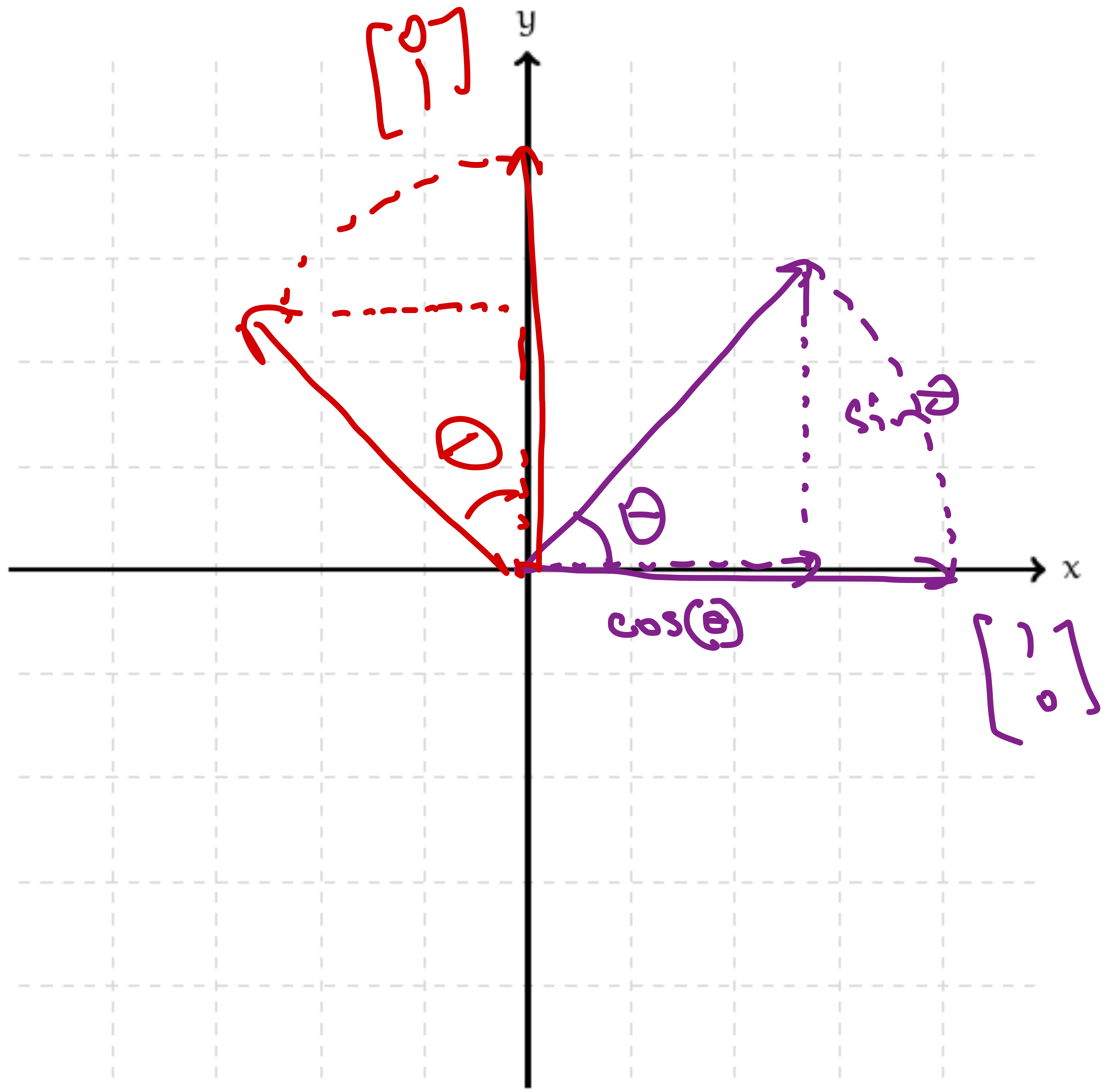
$$\begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix}$$

Geometry of Matrix Transformations

Matrix transformations change the "shape" of a set of set of vectors (points).

Example: General Rotation

How does rotation affect the standard basis?



$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

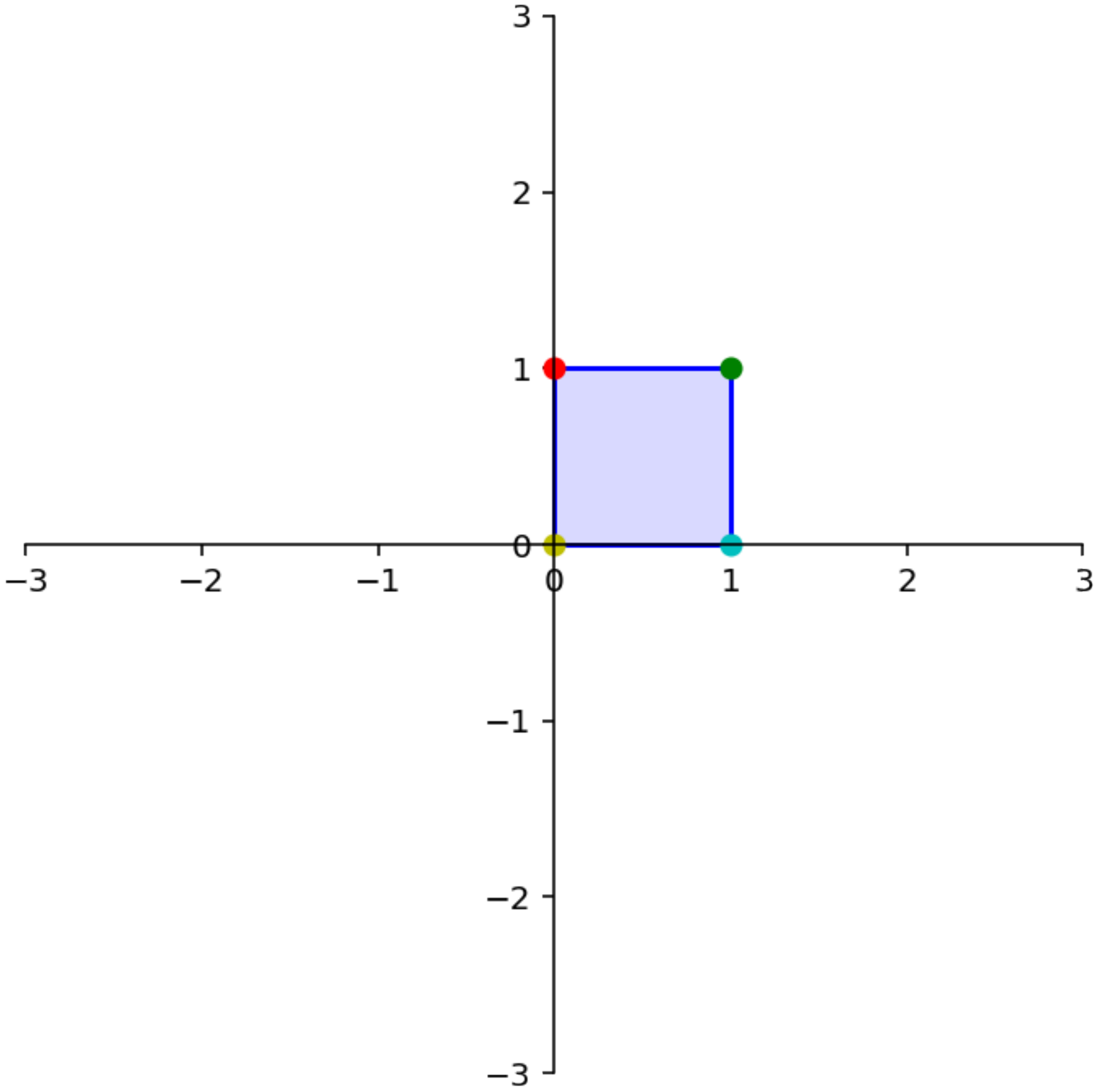
Rotation Matrix

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

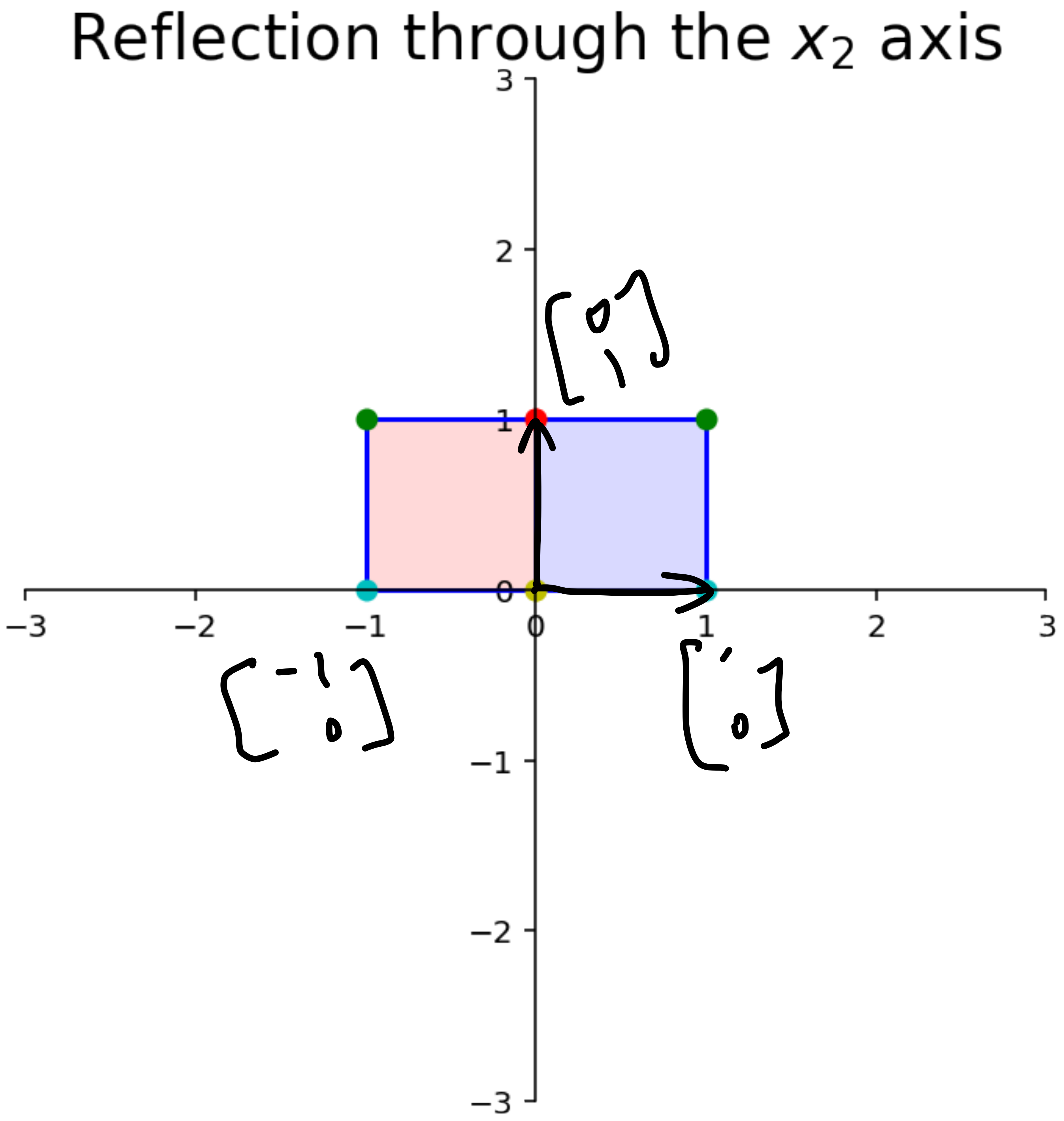
The Takeaway: We can figure out the matrix that implements a linear transformations by understanding what it does to the standard basis

The Unit Square

The ***unit square*** is the set of points in $\mathbb{R}^2$ enclosed by the points $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$.



Example: Reflection through the x_2-axis



$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

workshop